

Smarter Work, Safer AI:

A Report on Integrating Generative AI into Business Practices



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CSIT205

Generative AI

Assignment 2 Report

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Introduction to Generative AI and its Significance.

Generative AI has changed from a novel idea that many people believe as a short-term fad to something that is already being used in day-to-day working life, often faster than companies can plan for. That kind of speed matters because it changes how we should think about how generative AI is, what it can do, and why companies are paying close attention to it.

So, what is Generative AI? At its core, Generative AI (GenAI) refers to AI techniques that learn representations from data and then use those representations to generate new outputs. GenAI can be defined as generating “**brand-new, unique artifacts**” that resemble, but don’t just repeat, the same original dataset. This distinguishes GenAI from older AI approaches that are mostly used to sort data or predict outcomes. While those systems focus on analyzing information that already exists, GenAI creates entirely new digital content.

GenAI is incredibly flexible for its use cases as it generates all kinds of content. It generates texts, images, videos, audio, code for software, and many other uses. What makes it stand out is how we interact with it. Instead of needing to learn complex commands, you simply describe what you need. In this way, GenAI is less of a tool with a singular purpose but more of a multipurpose system that can support many tasks.

The significance of GenAI for companies lies not only in the content it can generate but because of how rapidly it has been integrated into daily life. Microsoft’s 2024 Work Trend Index shows that usage **nearly doubled in only six months**, with 75% of global professionals now using the technology. However, while employees are adopting these tools on their own, many companies still lack a clear roadmap to integrate that use across the organization. The challenge is scaling individual use into consistent organisational usage. The next sections review industry use, business impact, and practical considerations for adoption.

Case Studies

Across all industries Generative AI integration is being used to boost output, address issues, and open new business opportunities. These case studies enable us to observe how different types of organisations are using this technology in useful ways. In this, we will see how fashion, consumer goods and services all benefit from generative AI.

The New Jersey-based fashion company **Amarra** incorporates generative AI into its design process. AI helps in the creation of dress patterns, designs and marketing visuals. This offers customers more personalized options for their dresses. Fashion brands can use AI to stay creative and competitive by combining efficiency and creativity, which is shown in Amarra's experience (Hightower, 2025).

Starbucks, a global beverage retail company, uses generative AI (Deep Brew) to transform goods retail. Deep Brew AI personalises drink recommendations, boosting loyalty from the consumers and upselling. AI-driven R&D speeds up beverage innovation by analysing flavour trends and customer data. Marketing campaigns are localised and automated, improving engagement and digital sales. AI forecasts demand to reduce waste and optimize inventory. This enhances customer experience, efficiency and sustainability (Kotorchevikj, 2021)

Walgreens Boots Alliance is one of the largest retail pharmaceutical companies that operates thousands of outlets in the US, Europe and Asia. AI helps in using powered chatbots and answers customers' enquiries about medications, side effects, and purchase history. By doing this, not only does it reduce call centre workload but also improves accessibility and faster customer support. AI can be efficient and enhance trust in pharmacy services (Stapleton, 2022).

Across fashion, consumer goods and healthcare, generative AI is being very effective, efficient and innovative. But at the same time, businesses must be careful about issues like privacy breaches, fairness and ethics. Generative AI is becoming a powerful tool for shaping the future of business by helping companies work smarter and engage people deeply and establish themselves across industries worldwide.

In-Depth Analysis of the Impact on Industry

Improved Personalization: Generative AI helps stores improve the shopping experience for customers by making it more personalized as they provide tailored product recommendations, targeted promotions and customized marketing messages from the customer data. This helps to improve customer engagement, satisfaction and purchase conversion rates.

Operational Efficiency: AI helps with tasks that are done repeatedly such as crafting product description, generating promotional materials and even responding to customer service questions. This lowers the operational expenses and enables businesses to expand their tasks without increasing labor demands. AI can assist with repetitive tasks.

Enhanced Demand Forecasting: Generative AI looks at past sales data and consumer behaviour trends to produce more accurate predictions of the demand. This helps retailers in better control of their inventory and reducing stock shortages or excess inventory.

Transformation In Customer Service: These computer programs such as AI-driven chatbots and virtual assistants can respond to customer inquiries more quickly. Customers can request at any time of the day, 24/7 assistance, which can cut customer service costs.

Industry Obstacles: Even though Generative AI offers advantages, it can also cause certain issues. When we are making use of Generative AI may raise certain concerns. Such as data privacy, the possibility of inaccurate AI-information being shared and ethical implications. To prevent and deal with these issues, governance structures and human oversight is required to ensure appropriate usage.

Overall, Generative AI is changing the retail industry by delivering better customer experience, more work efficient, smarter decisions and enabling businesses to gain significant competitive advantages in the market.

Strategic Recommendations for Integration

The introduction of Generative AI is a paradigm shift in which businesses should recognize it as a key organizational skill rather than just a bunch of separate solutions.

McKinsey (2025) states that the **10-20-70 rule** is the most effective and efficient way to implement this transformation: **10% effort on algorithms, 20% on data/technology, and 70% on people and process transformation.**

To maintain surveillance from one centralized location, businesses should set up an AI Strategy & Governance Office. This body creates an AI Steering Committee to:

- Establish Acceptable Use Policies that protect intellectual property.
- Maintain an AI Risk Register to mitigate biases and preconceptions.
- Make sure that all tools and devices follow the "zero-trust" data standards (Gartner, 2026).

However, It's crucial to remember that **GenAI is only as good as the data it accesses.**

Organizations should prioritize the following things first:

- **Data Unification:** Eliminating barriers to create a unified cloud data platform.
- **Role-Based Access:** Make sure that models can only access data from authorized users.
- **Metadata Management:** Using active surveillance to make sure that LLMs are using data that is accurate and up to date.

At the same time, businesses should prioritize high value pilots, Strategic integration begins with use cases that are low-risk and high-impact to demonstrate ROI immediately:

- **Marketing:** Automated content and SEO optimization.
- **Operations:** Summarization of complex dashboards and reports.
- **Support:** Drafts made with AI assistance for others to review.

Furthermore to minimize reputational risk, a Human-in-the-Loop model is needed. AI should aid with writing and analyzing, but humans ought to verify high-stakes outputs (such price, legal, or consumer communication) to make sure they are both accurate and ethical (HBR, 2025).

Another key factor is that the "**skills gap**" is the most significant issue to AI scalability. Companies need to take part in AI literacy programs that focus on training staff members on how to communicate effectively with models as well as showing workers how to audit and verify AI-generated work. Adopting a phased approach reduces risk and increases the likelihood of successful AI adoption.

Phased Implementation Roadmap:

- **Phase 1 (Foundational):** Secure governance and data infrastructure.
- **Phase 2 (Pilot):** Deploy departmental agents (HR and Marketing).
- **Phase 3 (Agentic):** Develop self-sufficient systems that can handle multi-step processes

Finally, success must be calculated through clear financial and operational KPIs using the standard ROI formula:

$$ROI = \frac{Net\ Gains\ (Efficiency + Revenue) - Total\ Cost\ of\ Ownership}{Total\ Cost\ of\ Ownership}$$

Analysis of How Generative AI can Contribute to Achieving Specific UN SDGs.

SDGs is a worldwide goal that all countries are trying to achieve. Generative AI has been a big part of people's lives recently and it could definitely help one of the 17 SDGs one of which is the eight, **"Decent Work and Economic Growth"**. Without exploiting the use of AI by increasing the unemployment rate, there are a few ways Generative AI could help enhance efficiency and productivity. Repetitive tasks and supporting innovation that waste resources could be assisted with automated programs. It enables business to work efficiently by generating brainstorming ideas, or code more efficiently as an example. Using Generative AI does not mean completely replacing human skills. Instead of seeing it as an enemy, businesses should take this opportunity to take advantage of it and work cohesively with this powerful tool. Additionally, AI-driven platforms can help employees learn and improve skills for evolving projects ahead.

SDG 12: "Responsible Consumption and Production" is another one of the 17 that could be improved using Generative AI. This SDG focuses on improving resource efficiency, overproduction, and many others. Generative AI tools like those helping with analyzing trends, customer insights, sales, and others will ensure there are no unnecessary resources wasted. Excess inventory, packaging waste, and other issues can be avoided to a certain extent with the help of Generative AI.

A simple example of how AI is helping businesses is AI chatbots. This one tool helps both customers and businesses create a more productive and efficient environment. Instead of waiting on one another on calls or emails, these AI chatbots could help with simple, straightforward enquiries. Otherwise, the message will then be forwarded to people in charge of customer service. This system makes it much more systematic and functional and saves time for both sides.

Considerations for Data Management, Privacy, and Ethics in the use of Generative AI.

For successful adoption of Generative AI at RetailTech Innovators, it depends not only on the technical expertise, but also the rock-solid privacy protection, data management & ethical governance within the company and the wider landscape. Generative AI systems, by its nature, require access to large volumes of organizational and customer data, making data security and governance critical to minimizing operation and reputational risk factors.

To address data management first, RetailTech must address existing inefficiencies and risks associated with decentralized practices when it comes to customer data. The company *must* address the prior data breach, where customer service conversations were downloaded onto an employee's laptop, bringing to the forefront the importance of a centralized data infrastructure and strict access controls. Generative AI systems, especially ones with sensitive customer-relevant learned data, should only operate on secured, organization-approved platforms, supported by data minimization practices and security principles to ensure that only the necessary and relevant data is processed. Role-based access can be implemented to decrease the area of attack while audit trails can help to track when something goes wrong to prevent unauthorized use or extraction of sensitive information.

Secondly, **privacy**; An obvious key concern, given the personal nature of customer data used in online retail, which includes sensitive data like purchase histories, browsing behaviour, customer service interactions and the likes. The use of Generative AI must comply with data protection principles such as purpose limitation, informed consent and secure storage. Sensitive, directly traceable data needs to be anonymized/pseudonymized wherever it is possible to reduce the risk of identification or tracing. Transparency is also critical, as customers must be informed of when AI systems are involved in the decision making processes such as personalized recommendations

or automated customer support so as to not blindsides the customer-base as well as allow them to choose to use these systems responsibly.

Speaking of which, **Ethical considerations** are also something that needs to be kept in mind when it comes to the AI systems. These considerations will shape responsible Generative AI deployment and use, both in and out of the company. AI systems that are trained on historical retail data may inadvertently reinforce existing biases, leading to unweighed product recommendations, unethical pricing strategies or being uniquely manipulative when it comes to marketing practices. Over-reliance on automated systems might also reduce human judgement in areas that *do* require empathy and deeper contextual understanding, namely, customer service. To address these risks, the company should implement ethical AI governance baselines and frameworks that include bias testing and clear accountability from AI outcomes.

Overall, RetailTech Innovators need to be keenly and uniquely understanding when it comes to addressing data management, privacy and ethical considerations. It is essential for building trust with both customers and stakeholders, especially with the booming nature of AI as it is. By rigidifying responsible governance into the Generative AI strategy, RetailTech can reduce overall risk while ensuring that AI adoption aligns with the businesses long-term sustainability. And what is arguably more important, is that keeping in mind these things allow RetailTech to forge and strengthen trust with its customers.

Conclusion

Generative AI is already being used at work, and in many cases the adoption is happening before companies have proper processes in place. For RetailTech Innovators, that conflict matters. The report has shown that GenAI is not just another analytics tool; it generates new outputs from prompts, including text, images and code, so it can be applied across marketing, customer support and internal operations.

The case studies also show why companies are interested in it. GenAI can improve personalisation and speed up routine content and support work. But the risks are evident. In retail, the same systems can expose private customer information, reproduce bias in recommendations, or produce confident but incorrect answers. These problems get worse when staff use tools in an unregulated way without clear rules.

RetailTech Innovators' next step is to make adoption consistent. That means using approved systems, tightening access to customer data, and keeping audit trails so mistakes can be traced. Important and sensitive outputs still need human review, especially anything customer-facing or related to pricing and policy. If RetailTech Innovators treats GenAI as part of normal operations rather than a side experiment, it can scale the benefits without repeating past mistakes like decentralised data handling.

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